Module 6: Polygenic Risk Score Portability in Lung Adenocarcinoma (LUAD)

Data and Methods

I used PGS003393 (best_LUAD), a published lung adenocarcinoma polygenic risk score (PRS) developed using European-ancestry genetic association data (Namba et al., 2023). I analyzed real genotypes from 2,504 participants in the 1000 Genomes Project, covering five ancestry groups (1000 Genomes Project Consortium, 2015).

The published score contains 74 variants. I successfully matched 64 variants and excluded 10 because their strand orientation could not be confirmed safely. I calculated the PRS using the matched genotype dosages and published effect weights.

The genotypes and ancestry labels are real, but age and LUAD diagnosis labels are simulated. Therefore, the discrimination and calibration results demonstrate the workflow, not actual LUAD prediction accuracy.

1. Portability: Why can a European-derived score perform differently across ancestries?

A PRS developed using European-ancestry data may not perform equally well in other populations because of differences in allele frequencies and linkage disequilibrium (LD). Allele frequency means how common a genetic variant is in a population. LD describes how strongly nearby variants are inherited together.

A genetic marker associated with disease in Europeans may not track the same causal variant as closely in another ancestry group. As a result, the same PRS weights may not transfer equally well (Martin et al., 2019).

In my simulated LUAD analysis, the AUC was **0.605 in EUR, 0.629 in AFR, 0.686 in EAS, 0.662 in SAS, and 0.585 in AMR**. Interestingly, EUR did not have the highest AUC. This shows that my simulation does not demonstrate a consistent loss of accuracy outside Europe.

Because the disease labels were simulated, I cannot conclude which ancestry group would have the best real LUAD prediction accuracy.

2. One Threshold, Many Groups: Is a single top-10% cutoff fair?

My whole-cohort top-10% PRS cutoff was **1.321**. However, the percentage of participants above that cutoff differed across ancestry groups:

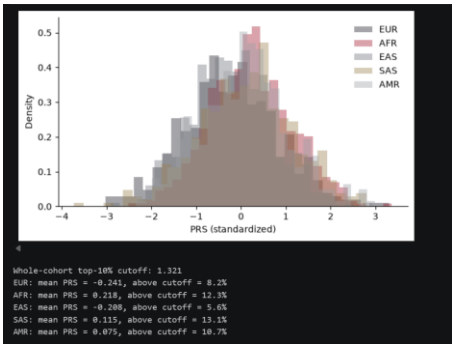
Ancestry Above cutoff

EUR	8.2%
AFR	12.3%
EAS	5.6%
SAS	13.1%
AMR	10.7%

This means a single cutoff does not select the same proportion of people from every group. For example, SAS had 13.1% above the threshold, while EAS had only 5.6%.

If a clinic used this cutoff unchanged, some ancestry groups would be flagged more often than others, even without evidence that the same score indicates the same clinical risk.

I would not consider this threshold ready for clinical use. It needs validation against real disease outcomes and calibration across populations before it can guide decisions (Martin et al., 2019).



3. Discrimination vs. Calibration: What did each tell me?

Overall AUC: 0.639

EUR: AUC = 0.605

AFR: AUC = 0.629

EAS: AUC = 0.686

SAS: AUC = 0.662

AMR: AUC = 0.585

Discrimination measures how well a model separates cases from controls. I evaluated it using AUC (area under the receiver operating characteristic curve). An AUC of 0.5 indicates chance-level discrimination, while 1.0 indicates perfect separation.

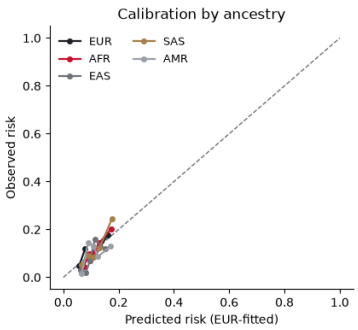
My overall PRS AUC was **0.639**. EAS had the highest ancestry-specific AUC at **0.686**, while AMR had the lowest at **0.585**.

Calibration measures whether predicted probabilities agree with observed outcome frequencies. I fitted a logistic regression model using the EUR group and applied it to all five groups.

My calibration graph showed differences between predicted and observed frequencies. Although EAS had the highest AUC, that does not mean its predicted probabilities were perfectly calibrated.

The simulated LUAD case rate was **10.5%**, with predicted probabilities ranging from **2.7% to 29.6%**.

This demonstrates an important difference: a model may rank people reasonably well but still provide inaccurate absolute risk estimates. Both discrimination and calibration must be assessed before clinical use.



4. Clinical Utility and Limitations

A validated LUAD PRS could potentially help identify people with higher inherited susceptibility and support research on personalized cancer prevention. It might eventually complement established risk factors such as age, smoking history, family history, and environmental exposures (Namba et al., 2023).

In my analysis, the five-fold cross-validated AUC was **0.609 for age alone** and **0.675 for age plus PRS**, an improvement of **0.066**.

However, this improvement does not demonstrate real clinical benefit because both age and LUAD outcomes were simulated, and the simulated outcome generation partly depended on the PRS.

I would not use these results to diagnose LUAD, recommend screening, or make treatment decisions.

Before clinical use, I would need real LUAD cases and controls, reliable clinical variables, independent validation across ancestry groups, and evidence that the score improves decisions beyond existing risk factors. I would also need to evaluate calibration, false-positive rates, and potential health disparities.

5. Artifact vs. Biology: Understanding the APOE Example

The original assignment discusses APOE ε4, an Alzheimer's disease risk allele. It describes a teaching simulation where a genetic marker linked to APOE ε4 is almost absent in the African-ancestry group, making the score perform near chance.

This is a property of the simulated dataset, not proof that APOE ε4 has no biological importance in people with African ancestry. A weak or missing genetic proxy can make a model appear inaccurate even when the underlying risk allele remains relevant. Differences in LD can affect how well a proxy represents a causal variant (Martin et al., 2019).

My analysis used LUAD rather than the original Alzheimer's dataset. Therefore, the APOE example is not part of my LUAD results, and my African-ancestry AUC was **0.629**, not near chance.

The main lesson is that poor performance in one dataset does not automatically mean a genetic variant has no biological effect. We must separate data limitations and modeling assumptions from genuine biological differences.

Proof of Real Data — Section 2

- **Source:** REAL
- **Participants:** 2,504
- **Published score size:** 74 variants
- **Variants used:** 64
- **Variants excluded:** 10 (strand ambiguity)
- **Ancestry counts:** AFR 661, AMR 347, EAS 504, EUR 503, SAS 489

File fingerprints (first eight SHA-256 characters):

- genotypes.csv: 62fbc28e
- variants.csv: 5349e113
- score.csv: e8d74cb7
- meta.csv: fcf43773

Note: Real 1000 Genomes genetic data were combined with simulated ages and LUAD outcome labels for this classroom analysis.

Conclusion

My LUAD analysis showed differences in PRS distributions, top-10% classification rates, and model performance across ancestry groups. Adding PRS to age increased the simulated AUC from 0.609 to 0.675.

However, **these results do not validate real LUAD prediction or ancestry portability**, because the disease outcomes were simulated. The main conclusion is that PRS calculations require careful allele matching, ancestry-specific evaluation, and independent clinical validation before they can be considered reliable or fair.

References

1000 Genomes Project Consortium. (2015). A global reference for human genetic variation. *Nature, 526*(7571), 68–74. https://doi.org/10.1038/nature15393

Martin, A. R., Kanai, M., Kamatani, Y., Okada, Y., Neale, B. M., & Daly, M. J. (2019). Clinical use of current polygenic risk scores may exacerbate health disparities. *Nature Genetics, 51*, 584–591. https://doi.org/10.1038/s41588-019-0379-x

Namba, S., et al. (2023). Common germline risk variants impact somatic alterations and clinical features across cancers. *Cancer Research, 83*(1), 20–27. https://doi.org/10.1158/0008-5472.CAN-22-1492

PGS Catalog. (2023). *PGS003393: best_LUAD* [Polygenic score]. https://www.pgscatalog.org/score/PGS003393/